

Story 6: Upgrade / downgrade route

6.1. Story:

As a customer support representative,

I want to assign a route to the shipment who match the ETA,

So that the order will be delivered to the customer within the promised ETA.

 The current basic (MVP) route assignment, already developed, remains in place. New upgrade logics are just an expansion of current functionality.

6.2. SCENARIO 1: Off season

6.2.1. Background

Off season upgrades and downgrades can run almost automatically, with minimum engagement of CSR. We assume a predictable number of shipments that will be upgraded within a predictable budget. This flow will be fully managed in the shipment service.

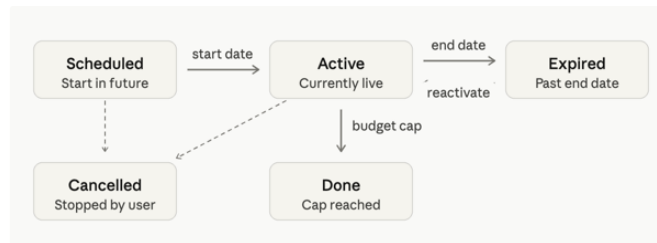
6.2.2. High level functionality

- There will be a UI, opened for CSR managers, where upgrade and downgrade rules can be configured. This will give us the exact conditions, when should we trigger the upgrade / downgrade flows.
- The upgrade / downgrade logic itself will run as an extension of current route assignment process, when the shipment is still on draft. The result of the process will be a ready to pack shipment with a route as close as possible to the order ETA. It also means that if for some reason the initial assign route process failed, we are not triggering any further logic.
- The domain event data of upgrade / downgrade will be used both by BI to control budgets and shipping cost reports, and also by Kustomer (or communication service) to trigger proper updates to the end customer. Because of that, it is crucial to fire a domain event at the end of the process.

6.2.3. The rule engine

- The rule engine will have a simple UI within the shipment service, on the same UI UX principles as the packing instructions and the shipment alerts. It will consist of two views:
 - List view showing the rules by
 - names
 - action (downgrade or upgrade)
 - status
 - budget and it's usage (in case of upgrade), and accumulated saved money (in case of downgrade)
 - how many times the rule applied while it (was) active
 - start and end day
 - Single view (drawer) where a rule can be added or edited
- Rules will work with an AND connector and will based on:

- Name (free text up to 50 ch, mandatory), uniqueness should be enforced between all rules in all statuses
- Brand (multiple selection) based on the shipment service brand configuration
- Event level (free field, option to enter several event levels). Event levels are not restricted and not integrated with shipment service (if feasible, should be restricted and synced once a day). So currently no validation is possible.
- Packing facility (multiple selection) based on the shipment service facility configuration
- Destination country (multiple selection, 2 letter code from country list)
- SKUs to include (free field, option to enter multiple SKUs). Product SKUs are not synced between the master catalog (or TG Supplier) and the shipment service, this will be done later in the scope of the factory catalog project. So currently no validation is possible.
- SKUs to exclude (same)
- Total order cost paid by the customer (check if we have this information in the shipment service?)
- Delivery timeframe: one of the following options must be selected
 - ETA threshold in days (digit)
 - *The meaning of thresholds: if ETA threshold is 2 and the original order ETA is May 10, the shipment should arrive between May 8 (including) and May 12 (including).*
 - Exact Delivery day (date picker)
 - *The meaning of exact delivery day: the shipment must be delivered until a specific day*
- Start day and expiration day (both mandatory): when the flow should move to status active and to status expired, with day picker. Logic will use shipment service server time.
 - If you don't enter a start day, the status will be active.
 - If you enter a start day, an end day will be mandatory, and must be after the start day.
- Action: downgrade, upgrade (radio button, mandatory)
- When upgrade selected, the upgrade cost can be controlled per shipment OR per rule (radio button, mandatory to chose and fill out one).
 - Per shipment: one of 2 possible rules must be filled in
 - max upgrade cost in USD per shipment between original method cost and upgraded flow cost
 - *Example: if the max upgrade is 3\$ and the original method cost is \$7 and the upgraded route cost is \$9.8, it will be upgraded. If the upgraded route cost is \$10.1, it won't be upgraded because the upgrade cost will exceed the max.*
 - max upgrade cost in USD per shipment between shipping revenue (paid by the customer) and actual cost
 - Per rule: enter a value in USD. Rule will be suspended if the total upgrade cost amount reach this threshold.
- When creating a new rule there should be an option to clone an existing one from a drop down of the rules (by name)
- Statuses
 - **scheduled** — start date in the future
 - **active** — within the start/expiration window, cost cap not yet hit
 - **expired** — past the expiration date (can be edited and reactivated by moving the expiration day to the future)
 - **done** — cost cap reached (terminal)
 - **cancelled** — moved to this status manually by the user from scheduled or active
 - we are not deleting rules to maintain data integrity



- To clarify: "Upgrade cost" = upgraded route cost – original route cost. One number per shipment. Stored on the shipment, sent to BI.
- To clarify: since we operate this feature in the shipment service, we are supporting split shipments out of the box. Each shipment is treated separately, and multiple shipments under one order can be upgraded or downgraded without dependency on other shipments under the same order.
- To clarify: about budget limitation: 2 budget gates can block an upgrade:
 - Per-rule budget — running total of upgrade costs against a cap. When the cap is reached, the rule moves to **done** and cannot be applied anymore.
 - Per-order budget — user picks one of two formulas:
 - A: $\text{upgraded cost} - \text{original cost} \leq \text{max}$
 - B: $\text{upgraded cost} - \text{shipping revenue} \leq \text{max}$
 - Only the selected option has to pass.
- Important but not in the scope of this story: shipments of zero value orders should be always excluded from any upgrade logic, based on reorder flag or total order paid by customer = 0. We have a dedicated ticket to ensure this, [SHP-577: Shipping reorders](#) **TO DO**

UI implementation:

<https://claude.ai/artifacts/latest/019e5a79-bad1-762f-aed5-502c892b54eb>

list view

Upgrade Rules

Configure upgrade and downgrade rules applied during next segment

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single view

- select upgrade or downgrade (mandatory)
- clone a previous rule
- name the rule (mandatory)
- delivery day condition (mandatory, either threshold or specific day)
- activation rules
- cost control (either per shipment or per rule)
- scheduling and status

New Rule

Action *

Upgrade

Find a faster route closer to ETA

Downgrade

Find a slower route within threshold

General

Clone from existing rule (optional)

Select a rule to clone...

Rule Name *

e.g. Standard upgrade — US Rings

0/50 — must be unique

Rule Builder

Delivery day conditions *

ETA threshold (days)

Specific delivery day

Choose one: an ETA tolerance window, or a fixed target delivery date. Only one can apply.

ETA Threshold *

=

2

Activation Logic

Brand

In

Select brands...

AND

Destination Country

In

Select countries...

AND

Event Level

In

Enter numbers 1-99 (comma-separated)

AND

Packing Facility

In

Select facilities...

AND

SKUs

In

Enter SKUs (comma-separated)

AND

SKUs

Not In

Enter SKUs to exclude...

AND

Total Order Value

Greater than or equal to

0

Cost Control

Per shipment

Per rule

Max upgrade cost *

USD 3

Per-shipment cap: upgraded cost - original cost ≤ max. Evaluated independently for each shipment.

Status & Effective Dates

Start Date *

dd/mm/yyyy

Expiration Date *

dd/mm/yyyy

Status

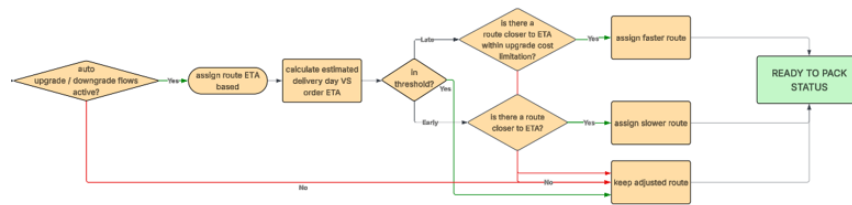
Active

Auto-set based on start date

New rule will be Scheduled.

Create Rule

6.2.4. Upgrade / downgrade logic



- After we run the existing route assignment process, if the process succeed, and there are active downgrade / upgrade rules, we check if the shipment qualifies for at least one rule of the same action.
 - I.e., the same shipment can be checked both for upgrade and downgrade rules, because we may need to either downgrade or upgrade it.
 - Obviously the same shipment cannot qualify both downgrade and upgrade rules, but it may qualify for multiple upgrade rules, for example. We will take the first rule it meets.
- If the shipment meets a rule, we calculate the estimated delivery day with the shipping carrier service type that was already assigned to the shipment, and compare it to the threshold of the rule.
- **Calculation of delivery days:**
 - current day
 - plus packing time frame based on shipping routes config considering facility working days in the week per facility configs
 - plus shipping time frame based on shipping routes considering shipping working days
- Possible results of delivery day calculation:
 - **On time shipment:** the delivery day estimation is in the threshold of the shipment data ETA as configured in the rule.
 - As a result → the originally assigned route will remain (we keep the outcome of the initial “MVP” route assign logic).
 - **Early shipment:** the delivery day estimation is before the threshold of the shipment data ETA as configured in the rule AND the rule is a downgrade flow: the downgrade process should be initiated.
 - Downgrade process: search for a route that match the ETA threshold as configured in the rule AND has a cheaper cost then the originally assigned route.
 - If such route found → assign that route.
 - If not → keep originally assigned route.
 - To clarify downgrade process: we are not interested in downgrading if the new method costs more then the original one. Also, we are not interested in reducing the timeframe between the order ETA and estimated delivery day (in case we were about to be early by 3 days but with the new method we are going to late 2 days). We only downgrade if we meet precisely the ETA threshold.
 - **Late shipment:** the delivery day estimation is after the threshold of the shipment data ETA AND the rule is an upgrade rule. Upgrade process should be initiated.
 - Upgrade process: search for a route that match the ETA threshold as configured in the rule.
 - If such route found and upgrade cost is below max. upgrade cost per order → assign that route and update the overall budget of the rule (if it's rule based) with the upgrade cost.

- → If after updating the budget, the rule reached it's limit, it should move to done status.
- If such route not found → keep originally assigned route.
- To clarify upgrade process: if the original method is late by 3 days and the new method is early by 2 days, and the ETA threshold is 2 days, we upgrade (within the budget limitation). In the same conditions, if the ETA threshold is 1 day, we don't upgrade.

6.2.5. Data

Every outcome of the assign route logic must be recorded either as a stand alone data linked to shipment, or part of the shipment data itself.

The shipment data scheme needs to be adjusted:

Group	Current field (JSON)	Action	New field name	Condition / notes
Initial (baseline after MVP assign process)	original_route	Rename	initial_route	route assigned by MVP process — rename preserves the value
	estimated_delivery_date	Rename	initial_eta	est. delivery day of the initial route
	—	Add	initial_cost	cost of the initial route (USD)
Upgrade	is_upgrade	Keep	is_upgrade	boolean; true ⇒ upgraded route is the final one
	upgrade_reason	Keep	upgrade_reason	rule applied
	suggested_route	Delete → Add	upgraded_route	route after upgrade; only when is_upgrade
	suggested_upgrade_cost	Delete → Add	upgrade_cost	= actual_shipment_cost -

				<code>initial_cost</code> ; only when <code>is_upgraded</code>
	—	Add	<code>upgraded_eta</code>	est. delivery day of the upgraded route
Downgrade	<code>is_downgrade</code>	Keep	<code>is_downgrade</code>	boolean; <code>true</code> $\Rightarrow$ downgraded route is the final one
	<code>downgrade_reason</code>	Keep	<code>downgrade_reason</code>	rule applied
	—	Add	<code>downgraded_route</code>	route after downgrade; only when <code>is_downgrade</code>
	—	Add	<code>downgrade_saving</code>	= <code>initial_cost</code> - <code>actual_shipment_cost</code> ; only when <code>is_downgrade</code>
	—	Add	<code>downgrade_eta</code>	est. delivery day of the downgraded route
Order info	—	Add	<code>order_eta</code>	the promised order ETA
	<code>original_shipment_price_amount</code>	Rename	<code>order_shipment_price_amount</code>	shipment cost paid by the customer
	<code>original_shipment_</code>	Rename	<code>order_shipment_pri</code>	the only currency

	price_currency		ce_currency	field kept
Actual / final	actual_shipment_cost_amount	Rename	actual_shipment_cost	final route cost (USD, single value)
	actual_shipment_cost_currency	Delete	—	route costs are always USD
	suggested_shipment_cost	Delete	—	no replacement; actual_shipment_cost is the final cost

UI representation in the SHIPMENT UI:

List under the cost summary the `order_shipment_price_amount` and currency, and the `actual_shipment_cost`.

If there was an upgrade or downgrade, also indicate `is_upgrade` or `is_downgrade` together with the `upgrade_reason` or the `downgrade_reason` as well as the `initial_cost`, and the calculated `upgrade_cost` or `downgrade_saving`.

List under the shipping information the `order_eta` and the `estimated_delivery_date`

List under the shipping history audit on the shipping UI drawer and packing screen UI the following:

- Shipment was upgraded from `original_route` with `initial_eta` to `upgraded_route` with `upgraded_eta` triggered by `upgrade_reason`.
- Shipment was downgraded from `original_route` with `initial_eta` to `downgraded_route` with `downgraded_eta` triggered by `downgrade_reason`.

Shipment Full Details

Order ID: 400053363

Status: OnHold

Timeline

OnHold

29/05/2026 at 18:32

Order Created

29/05/2026 at 18:30

Shipping Information

Shipment ID:

SHP-688

Tracking ID:

-

Shipment Collection ID:

-

Packing Facility:

Kiryat Gat

Destination:

Manchester, GB

Carrier:

-

Cost Summary

Original Shipment Cost:

-

Actual Shipment Cost:

-

Suggested Cost:

-

Customer

Name:

Andy Davies

Phone:

+441613334455

Address:

456 Deansgate, Manchester, M1 2GS, GB

Also, a domain event need to be fired, that other systems can consume to make appropriate updates.

The new data must include (expand the common domain event scheme):

- whether the outcome was downgrade / upgrade / no change
- the rule that the shipment qualified, including it's name, ID and action (downgrade or upgrade)
- assigned route (by the initial "MVP" route assignment process): it's name, cost and delivery day estimation
- actual route (by extended upgrade / downgrade process): it name, cost and delivery day estimation and difference from the original price

6.3. SCENARIO 2: Peak season

6.3.1. Background

During peak season (Black Friday, Christmas, Mother's Day), we are not just having a significantly larger amount of orders a day, but also a higher customer sensitivity for ETA. The product, usually a gift, is expected to arrive until the holiday (Christmas or Mother's Day).

As a result, it's hard to anticipate how many orders will required update, how much it will cost. Also, it's crucial to update the customer. Since shipment is created upon "sent" and we are talking about orders that are "late supplier", and we want to notify the customers even before shipment creation that the order is late but we are doing our best to deliver on time, this feature needs to work from OCS (even before the shipment was created) with an API integration with the shipment service.

6.3.2. High level functionality

CSR agent will run a query on "late orders" within OCS by some filters (late by X days, brand, destination country etc). Once he has a list of orders that supposed to get upgraded in order to avoid being late, the shipment service will provide upgrade offerings by a new, designated API endpoint. CSR agent will see the options and based on budget consideration will approve or dismiss the upgrade suggestions.

❓ Investigate whether "late supplier" flag can be used to simplify the late order detection, or perhaps better to build this new logic.

Split orders: since this feature works in the scope of OCS and on an order level, split orders are not supported. We will treat each order based on the projected shipment with the latest estimated delivery day, which represents the bottleneck of the order from CSR perspective. If the bottleneck part of the order requires upgrade, the entire order will be upgraded (known limitation).

6.3.3. CSR "late orders" in OCS

- We are adding a new, designated report in OCS in which CSR with appropriate permission can run a report on late orders. To run a report, select filters:
- Main filtering: by late days (selection of numbers from 1 to 21). Late days are calculated by remaining production and shipping days (production started domain event timestamp + production days + packing day + shipping time frame of the original shipping route), compared to the current day and original ETA.
 - *For example: order with ETA 27.5.26 with product "X" received production started on 20.5.26. Product "X" has 3 production days, 1 packing day, and the assigned route has 7 days of shipping time frame. Estimated delivery is $20+3+1+7 = 2.6.26$ (including packing facility working days and shipping_working_days of the shipping route based on our configurations). Current day is 28.5. As a result, order is late by 5 days.*
- Additional filtering:
 - brand (multiple selection)
 - event level (multiple selection, can be restricted based on active event levels)
 - destination country (2 letter, restricted by country list)
 - supplier ID (multiple selection): need to be based on supplier ID from TG Supplier "production started" domain event. I.e., orders that were not assigned to any supplier, will not be part of any filtering.
 - SKUs included (check if possible to restrict in OCS)
 - SKUs excluded (check if possible to restrict in OCS)
- Upon filtering, the relevant orders are selected, showing
 - late days
 - ETA
 - estimated packing day
 - estimated delivery day based on the calculated above
 - order ID
 - destination country

- original shipping carrier service type
- brand
- event level
- supplier ID
- SKUs in the order
- Split orders: orders that received multiple “production started” events, with multiple suppliers (potentially split orders) - we only ask for upgrade suggestions based on the latest estimated delivery day (the projected shipment that will arrive last). In that way eventually the CSR will base upgrade decision relying on the bottleneck part of the order which will ensure customer satisfaction.

UI implementation before API

list view before API

Late Days	Order ID	ETA	Est. Packing	Est. Delivery	Country	Carrier Service	Brand	Event Level	Supplier	SKUs
Get late	ORD-784321	2028-05-27	2028-05-28	2028-06-02	US	USPS_PRIORITY	Myka	mothers_day	113	90-443, 90-447
Get late	ORD-784422	2028-05-25	2028-05-26	2028-06-04	CA	GLOBALPOST_ECONOMY	Myka	mothers_day	125	90-220
Get late	ORD-784518	2028-05-22	2028-05-25	2028-06-01	AU	DHL_EXPRESS	BrandB	mothers_day	122	NA-100, NA-101
Get late	ORD-784624	2028-05-30	2028-05-28	2028-06-03	UK	WALDO	Myka	mothers_day	118	88-001
Get late	ORD-784733	2028-05-28	2028-05-26	2028-06-02	US	USPS_PRIORITY	Myka	mothers_day	113	90-443

6.3.4. API to shipment service

- “Get upgrade options” API will be triggered by the CSR user in the UI, once the filtering resulted with a list of at least 1 order.
- API call will include
 - ORDER IDs
 - destination country
 - ETA
 - current shipping carrier service type
 - estimated packing day
 - estimated delivery day
 - supplier ID.
- Shipment service will
 - map for each order the supplier IDs to packing facility ID
 - find for each order the cost of the original route (based on the original carrier service type service level, just like in the regular route assignment logic)
 - find for each order, up to 3 suggested routes from the particular packing facility to the particular destination country, that are the closer to the ETA then the estimated delivery day

- find for each suggested such route the ETA (when the shipment can be delivered based on the packing day provided by OCS + shipping time frame)
- find for each suggested route the cost of that route
- As a result, the API respond will include for each order:
 - status, if upgrade options found or not
 - original carrier service type cost
 - suggested carrier service type name
 - suggested carrier service type cost
 - suggested carrier service type delivery day
- OCS will add and present the suggested upgrades, if found, in the same late order UI
- For each upgrade suggestion under each order, there is a checkbox to select. The suggestion with the closest ETA will be preselected
- On top of the list view, all possible upgrades will grouped to cards based on carrier service types that the routes belong to. When you select a certain card, all the suggested routes belong to the same carrier service type will be selected, if exist, under each order
- Each card will show the aggregated upgrade cost for the selected upgrade suggestions that belong to the same carrier service type
- “Approve” will initiate bulk upgrade process on the selected orders → carrier service type will change to the the new one within OCS.
- Clarification: it's possible that the suggested upgrades are **closer** to the original ETA then the original estimated delivery day, but still “far” from it. I.e., the order is so late that even with the fastest method, delivery will be (as an example) 5 days **after** original ETA. This is why these are only suggestions, and the CSR will have to make a decision if the upgrade worth is or not. It's up to their consideration if such case deserve an upgrade or not.

OCS post orders to shipment service to get potential upgrade suggestions (note the second ordered with 2 potential shipments):

```

1  {
2    "orders": [
3      {
4        "order_id": "5123467",
5        "destination_country": "US",
6        "eta": "2026-05-27",
7        "current_carrier_service_type": "USPS",
8        "estimated_packing_day": "2026-05-25",
9        "estimated_delivery_day": "2026-06-02",
10       "supplier_id": "113"
11     },
12     {
13       "order_id": "272536478",
14       "destination_country": "CA",
15       "eta": "2026-05-25",
16       "current_carrier_service_type": "YunExpress",
17       "estimated_packing_day": "2026-05-26",
18       "estimated_delivery_day": "2026-06-04",
19       "supplier_id": "125"
20     },
21     {
22       "order_id": "27859270",
23       "destination_country": "UK",
24       "eta": "2026-05-22",
25       "current_carrier_service_type": "Maillog",
26       "estimated_packing_day": "2026-05-25",
27       "estimated_delivery_day": "2026-06-01",
28       "supplier_id": "122"
29     }
30   ]

```

response of shipment service with upgrade suggestions (note the second order receiving independent upgrade options for each projected shipment).

```

1  {
2    "orders": [
3      {
4        "order_id": "5123467",
5        "status": "options_found",
6        "supplier_id": "113",
7        "packing_facility_id": "NZ",
8        "original_carrier_service_type": "USPS_PRIORITY",
9        "original_carrier_service_type_cost": { "amount": 7.50,
10       "currency": "USD" },
11       "upgrade_options": [
12         {
13           "carrier_service_type": "FEDEX_2DAY",
14           "cost": { "amount": 12.40, "currency": "USD" },
15           "upgrade_cost": { "amount": 4.90, "currency": "USD" },
16           "delivery_day": "2026-05-29"
17         },
18         {
19           "carrier_service_type": "FEDEX_STANDARD",
20           "cost": { "amount": 14.20, "currency": "USD" },
21           "upgrade_cost": { "amount": 6.70, "currency": "USD" },
22           "delivery_day": "2026-05-28"
23         },
24         {
25           "carrier_service_type": "UPS_NEXT_DAY",
26           "cost": { "amount": 22.80, "currency": "USD" },
27           "upgrade_cost": { "amount": 15.30, "currency": "USD" },
28           "delivery_day": "2026-05-27"
29         }
30       ]
31     },
32     {
33       "order_id": "272536478",
34       "status": "options_found",
35       "supplier_id": "125",
36       "packing_facility_id": "TH",
37       "other_supplier_ids": ["131"],
38       "original_carrier_service_type": "GLOBALPOST_ECONOMY",
39       "original_carrier_service_type_cost": { "amount": 9.20,
40       "currency": "USD" },
41       "upgrade_options": [
42         {
43           "carrier_service_type": "DHL_EXPRESS",
44           "cost": { "amount": 18.40, "currency": "USD" },
45           "upgrade_cost": { "amount": 9.20, "currency": "USD" },
46           "delivery_day": "2026-05-26"
47         },
48         {
49           "carrier_service_type": "FEDEX_INTL_ECONOMY",
50           "cost": { "amount": 15.60, "currency": "USD" },
51           "upgrade_cost": { "amount": 6.40, "currency": "USD" },
52           "delivery_day": "2026-05-28"
53         },
54         {
55           "carrier_service_type": "UPS_WORLDWIDE_SAVER",
56           "cost": { "amount": 20.10, "currency": "USD" },
57           "upgrade_cost": { "amount": 10.90, "currency": "USD" },
58           "delivery_day": "2026-05-25"
59         }
60       ]
61     },
62     {
63       "order_id": "27859270",
64       "status": "no_options_available",
65       "reason": "No faster route available from TG_FLORIDA to AU",
66       "supplier_id": "122",
67       "packing_facility_id": "HU",
68       "original_carrier_service_type": "DHL_EXPRESS",
69       "original_carrier_service_type_cost": { "amount": 22.80,
70       "currency": "USD" },
71       "upgrade_options": []
72     }
73   ]
74 }

```

UI implementation after API call

list view after API call

6.3.5. Domain event

- Once carrier service type is replaced in OCS, an order update domain event will be fired with the new carrier service type.
- If there is no shipment yet, the shipment will be created with the upgraded shipping carrier service type
- If the shipment already exists, the domain event will lead to the update of the shipping carrier service type in the shipment. It will be seamless from packers perspective.
- BI should also consume the event in order to track order upgrades
- Domain event must include:
 - original route as dictated by the cart / OCS: it's name, cost and ETA
 - upgraded route (the approved upgrade option): it's name, cost and delivery day estimation and difference from the original price
 - order information including brand, event level and destination country
 - reason for upgrade: "peak season OCS update"